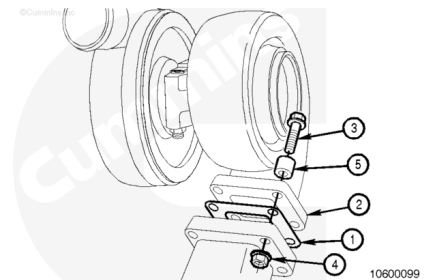


Trainings ▾

General Information

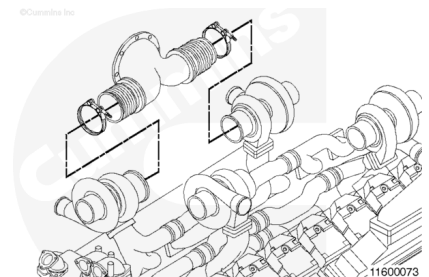
QSK45 Engines

The QSK45 engine exhaust manifold is designed so that all six cylinders on each bank are connected to a single common outlet located near the center, between number 3 and number 4 cylinders.



QSK60 Single Stage Industrial and Power Generation

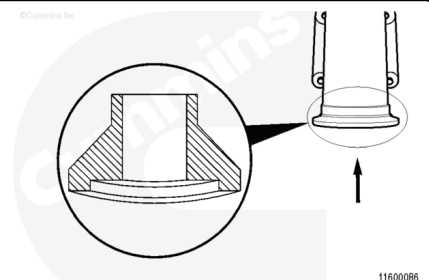
The QSK60 engine exhaust manifold is designed as a pulse-style with twin high-pressure turbocharging on each bank. Each turbocharger serves four cylinders. The turbocharger exhaust outlet feeds into a main exhaust outlet for each bank.



QSK60 Marine Applications

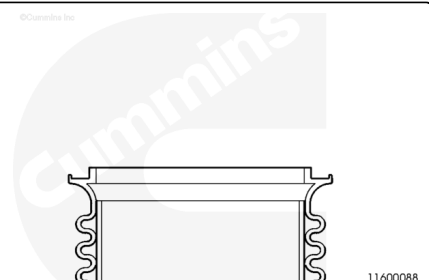
The marine exhaust system used on the QSK60 marine engine is designed with unique flanges.

The marine exhaust manifold side profile is shown in the illustration.



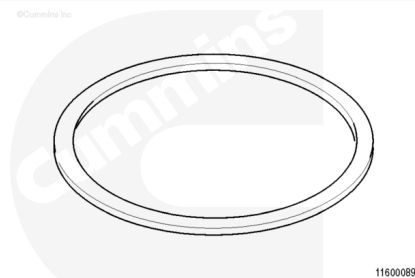
The marine exhaust manifold bellows has a different design than the industrial exhaust manifold bellows.

The marine exhaust manifold bellows is shown in the illustration.

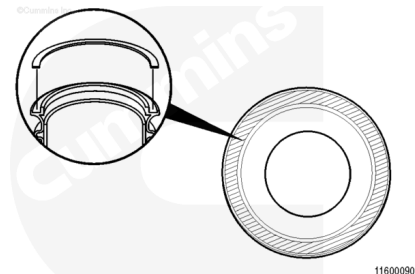


The exhaust manifold bellows gasket is located in the end of the bellows.

The bellows will **not** seal without the gasket in place.



The illustration shows the marine exhaust manifold bellows with the gasket fitted in place.



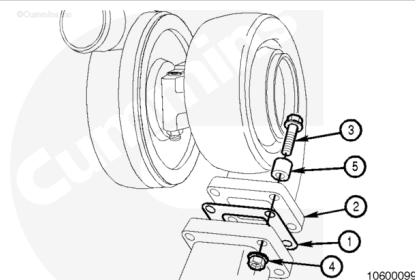
Preparatory Steps

QSK45 Engines

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

Remove the two turbochargers. Refer to Procedure 010-034 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>)

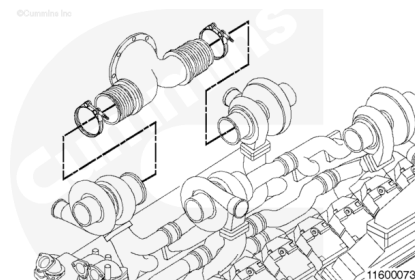


QSK60 Single Stage Industrial and Power Generation

⚠ WARNING ⚠

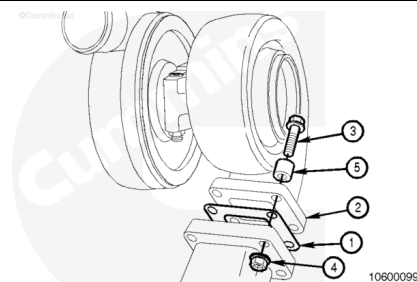
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Remove the four clamps and the exhaust connection.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

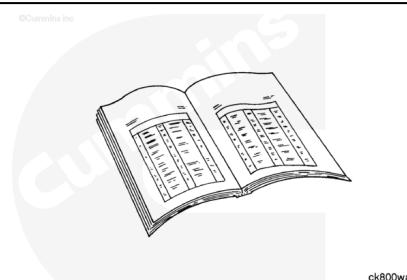


Remove the four turbochargers. Refer to Procedure 010-034 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>)

QSK60 Marine Applications

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

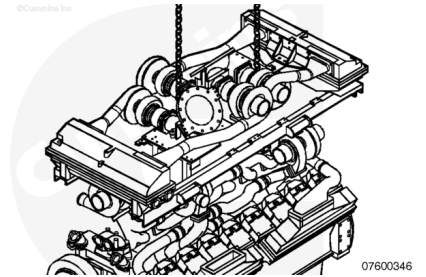


- Remove the aftercooler heat shields. Refer to Procedure 010-129 in Section 10.
(</qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html>)
- Remove the turbocharger inlet connections. Refer to Procedure 010-130 in Section 10.
(</qs3/pubsys2/xml/en/procedures/56/56-010-130-tr.html>)
- Remove the turbocharger heat shields. Refer to Procedure 010-076 in Section 10.
(</qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html>)
- Remove the four clamps and the exhaust connection.
- Remove the four turbochargers. Refer to Procedure 010-034 in Section 10.
(</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>)
- Remove the exhaust manifold heat shields. Refer to Procedure 011-015 in Section 11.
(</qs3/pubsys2/xml/en/procedures/56/56-011-015-tr.html>)

QSK60 Two Stage

⚠ WARNING ⚠

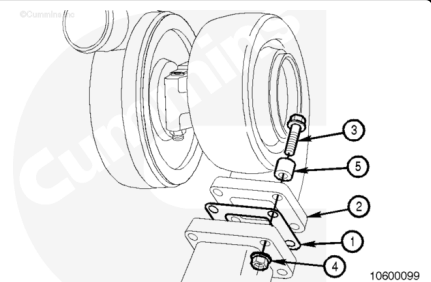
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the two-stage platform. Refer to Procedure 010-082 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-082-tr.html>)

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



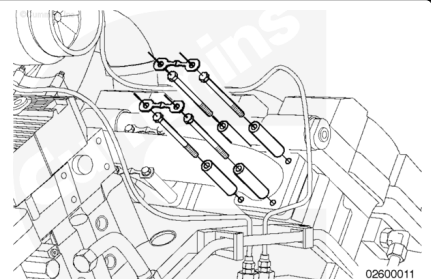
Remove the high-pressure turbochargers. Refer to Procedure 010-034 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>)

Remove

Industrial and Power Generation

⚠ CAUTION ⚠

The exhaust manifolds must be disassembled to remove them from the engine. Failure to disassemble the exhaust manifolds will result in exhaust manifold bellows damage.



When removing the exhaust manifolds, remove one section of the exhaust manifold at a time, beginning from either end of the manifold. Note the location and orientation of each section during removal.

Remove the exhaust manifold mounting capscrew lock tabs. Discard the lock tabs.

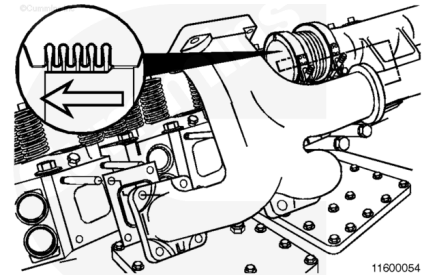
Remove the exhaust manifold mounting capscrews and spacers.

Note the direction of the bellows. They **must** be installed correctly for proper engine performance. If a directional arrow is **not** visible, mark the bellows. The open end of the sleeve inside the bellows **always** points in the direction of exhaust flow (toward the turbine inlet).

Remove the V-band clamp from either end of the bellows.

Remove the end section of the exhaust manifold.

Discard the manifold gaskets.



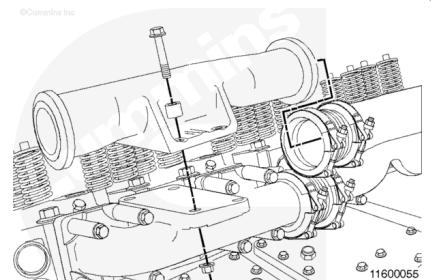
⚠ CAUTION ⚠

Never attempt to compress the exhaust manifold bellows. Bellows compression can result in damage to the bellows and an exhaust leak.

Note : It is **not** always necessary to remove all exhaust manifold sections and bellows. Remove **only** those components required.

Remove the V-band clamp and the bellows section from the next section of the exhaust manifold.

Repeat the steps above until the required sections of the exhaust manifold and bellows have been removed.



QSK60 Marine Applications

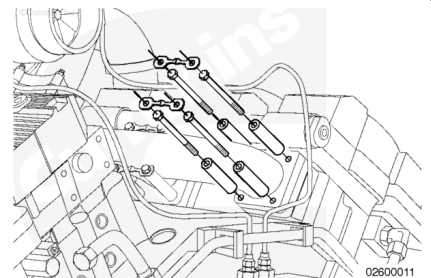
⚠ CAUTION ⚠

The exhaust manifolds must be disassembled to remove them from the engine. Failure to disassemble the exhaust manifolds will result in exhaust manifold bellows damage.

When removing the exhaust manifolds, remove one section of the exhaust manifold at a time, beginning from either end of the manifold. Note the location and orientation of each section during removal.

Remove the exhaust manifold mounting capscrew lock tabs.

Discard the lock tabs.



Remove the exhaust manifold mounting capscrews and spacers.

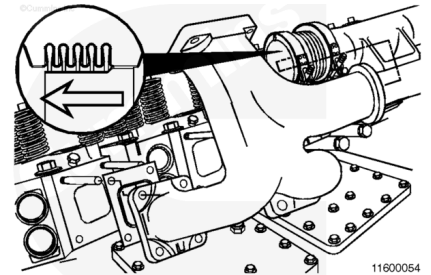
Note the direction of the bellows. They **must** be installed correctly for proper engine performance. If a directional arrow is **not** visible, mark the bellows. The open end of the sleeve inside the bellows **always** points in the direction of exhaust flow (toward the turbine inlet).

Remove the V-band clamp from either end of the bellows.

Remove the end section of the exhaust manifold.

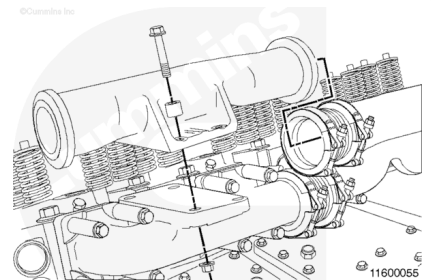
Remove the exhaust manifold bellows gaskets located on either side of the bellows.

Discard the manifold gaskets.



⚠ CAUTION ⚠

Never attempt to compress the exhaust manifold bellows. Bellows compression can result in damage to the bellows and an exhaust leak.



Note : It is **not always** necessary to remove all exhaust manifold sections and bellows. Remove **only** those components required.

Remove the V-band clamp and the bellows section from the next section of the exhaust manifold.

Remove the exhaust manifold bellows gaskets located on either side of the bellows.

Repeat the steps above until the required sections of the exhaust manifold and bellows have been removed.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause personal injury.

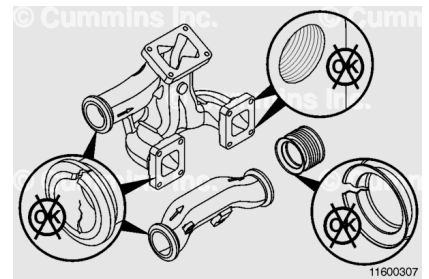
Use a steam cleaner to clean the exhaust manifold and components.

Inspect the exhaust manifold components for cracks, warpage, capscrew marks in the manifold flange holes, and leaks at the joints.

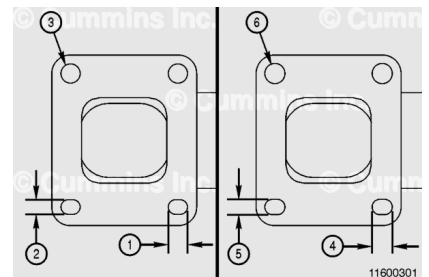
Inspect the bellows for cracks or other damage.

If any of the sections leak or are damaged, the manifold section **must** be replaced.

If capscrew thread marks are visible in the manifold flange holes, rework the holes and slots to new dimensions.



Note : In old manifolds, the flange holes have a 13.5 mm [0.531 in] diameter and the slots are 13.5 x 10.25 mm [0.531 x 0.4035 in]. In new manifolds, the flange holes have been increased to 14 mm [0.551 in] diameter and the slots to 14 x 10.5 mm [0.551 x 0.4133 in]. The holes and slots center axes remain in the same positions as in the previous designs. If there is capscrew damage or capscrew replacement, the holes and slots in the old manifolds **must** be machined to the dimensions of the new manifolds. The increased clearance between the capscrews and the holes and slots in manifolds will reduce the capscrew load due to thermal expansion of the manifold during engine operation.



Old Exhaust Manifolds

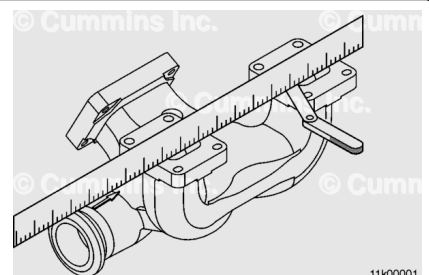
1. 13.5 [0.4035 in]
2. 10.25 mm [0.4035 in]
3. 13.5 mm [0.531 in]

New Exhaust Manifolds

4. 14 mm [0.551 in]
5. 10.5 mm [0.4133 in]
6. 14 mm [0.551 in]

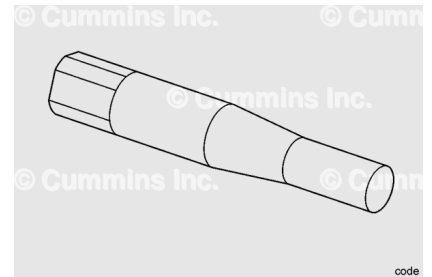
Place a straight rule, or equivalent, over the contact surface of each the exhaust manifolds as shown in the illustration.

Use feeler gauges to measure if there is any gap between the straight edge and the contact surface of the manifold. If the gap is greater than 0.2 mm [0.0078 in], it is **not** to be reused.



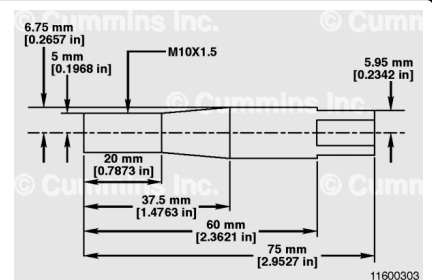
Install

A manifold-centering dowel tool should be fabricated locally to assist in the installation of the gaskets and the exhaust manifold sections.



Use the dimensioned illustration to fabricate four manifold centering dowels from a length of 14 mm [0.5511 in] diameter standard steel bar. The tolerance for dimensions in the illustration is ± 0.1 mm [± 0.0039 in].

This tool is intended for manual use.



Industrial and Power Generation

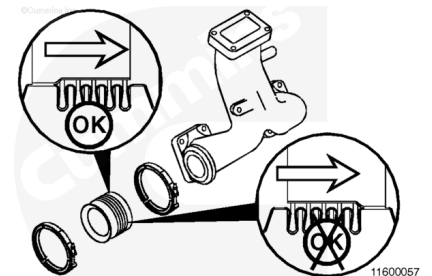
Note : The exhaust manifold for the QSK60 is shown. The procedures for the QSK45 are similar.

The exhaust manifold **must** be assembled as it is being installed.

Do **not** tighten the bellows clamps until all the exhaust manifolds are installed.

Apply anti-sieze compound, Part Number 3824879, to the exhaust mounting capscrews.

The exhaust manifold bellows are directional. Make sure they are installed in the proper manner when installing the exhaust manifold.



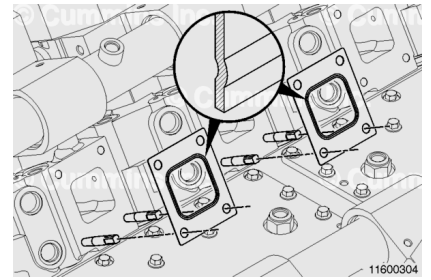
Ensure that the threads in the cylinder heads are free from any debris. If needed, the threads may be cleaned by tapping out the thread.

All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 5 and 6, or left bank cylinder heads number 3 and 4.

Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.



Place the exhaust manifold section onto the centering dowels.

Note there are two different capscrews available for use; a stainless steel capscrew and a steel capscrew. The stainless steel capscrew can be identified as it will have the part number on the capscrew head. The stainless steel capscrew **must** also be fitted with the new spacers, these can be identified as they have two grooves around the circumference.

For the stainless steel capscrew, lubricate the threads and the capscrew underhead with Loctite® LB8009 anti-seize lubricant. Do **not** use copper grease to lubricate the stainless steel capscrew.

For the steel capscrew, lubricate the threads and capscrew underhead with copper grease.

Install capscrews and spacers into the upper holes and center the manifold section.

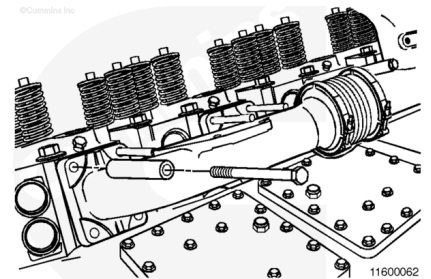
Tighten the upper capscrews.

Torque Value: 10 n•m [89 in-lb]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

Tighten all eight capscrews in the exhaust manifold section.

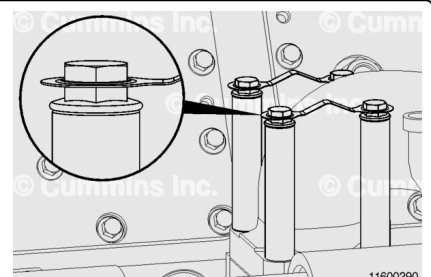
Torque Value: 47 n•m [35 ft-lb]



⚠ CAUTION ⚠

Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.

Note : Install the lockplate approximately one half of the height of the capscrew head. Installing the



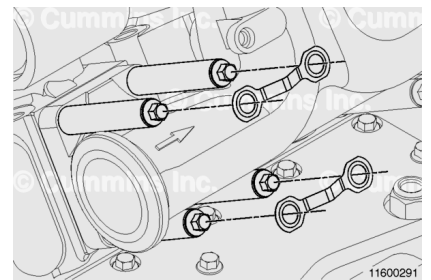
lockplate until it bottoms can cause it to break.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

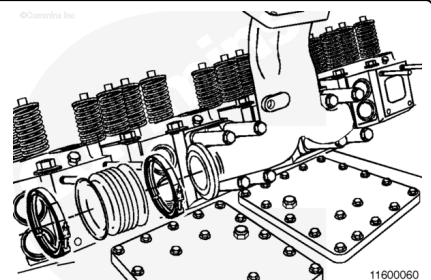
Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.



Make sure the bellows are installed in the proper direction.

All of the bellows clamps **must** be installed with the clamp screw toward the opposite bank of the engine, with the screw pointing up.

Loosely install the bellows clamp and the bellows on the manifold section just installed.

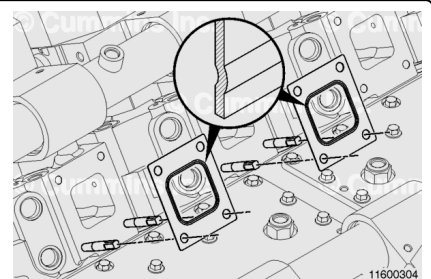


All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 3 and 4, or left bank cylinder heads number 5 and 6.

Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.

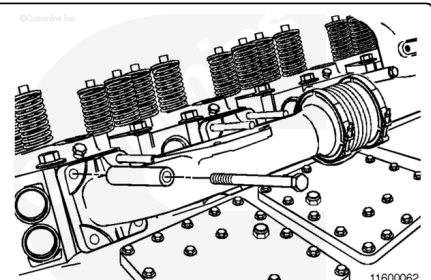


Place the exhaust manifold section onto the centering dowels.

Install capscrews and spacers into the upper holes and center the manifold section.

Tighten the upper capscrews.

Torque Value: 10 n•m [89 in-lb]



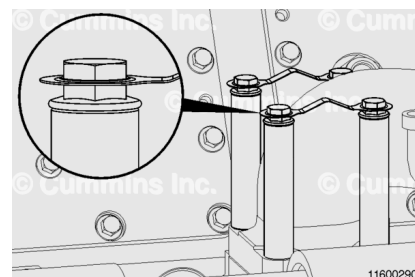
Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

Tighten all eight capscrews in the exhaust manifold section.

Torque Value: 47 n•m [35 ft-lb]

⚠ CAUTION ⚠

Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.



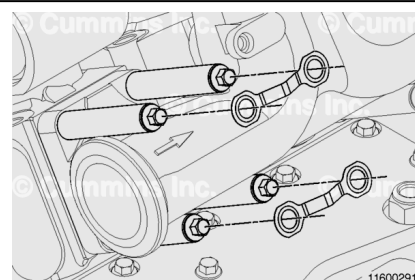
Note : Install the lockplate approximately one half of the height of the capscrew head. Installing the lockplate until it bottoms can cause it to break.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.



Install the bellows clamp.

Tighten the bellows clamps.

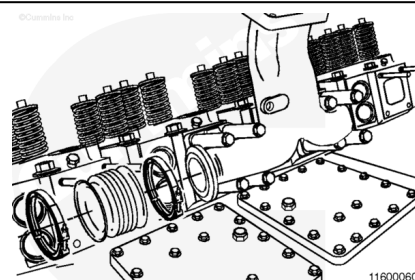
Note : There are two types of V-band clamps on the QSK45 and QSK60. They can be identified by the nut size.

Torque Value:

8mm (5/16 inch): 17 n•m [150 in-lb]

Torque Value:

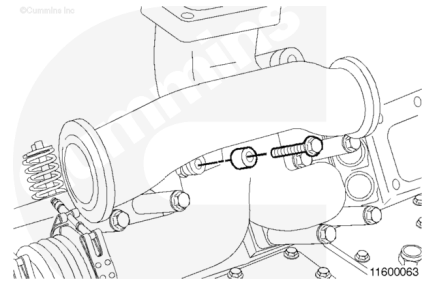
6mm (1/4 inch): 10 n•m [89 in-lb]



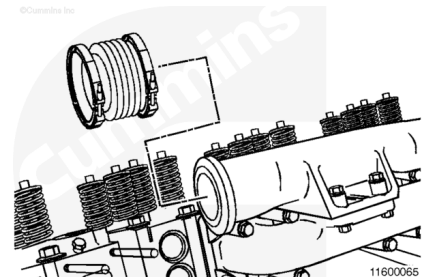
Ring dowels are used to locate and position the crossover section.

Position the crossover exhaust manifold section to the exhaust manifold section for right bank cylinders number 7 and 8, or left bank cylinders number 1 and 2.

Do **not** tighten the mounting capscrews.



Loosely install the bellows clamps and bellows to the ends of the upper exhaust manifold section.

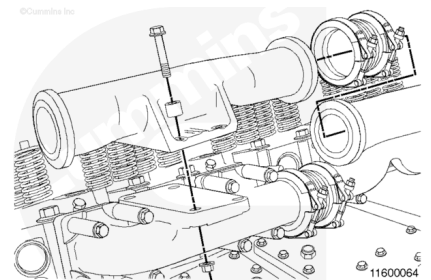


Position the crossover exhaust manifold section on top of the exhaust manifold section for right bank cylinders number 3 and 4, or left bank number 5 and 6.

Do **not** tighten the mounting capscrews.

Loosely install the bellows clamps and bellows to the upper exhaust manifold section mounted to the turbocharger mounting exhaust manifold section for right bank cylinders number 5 and 6, left bank cylinders number 3 and 4.

Do **not** tighten the bellows clamps.

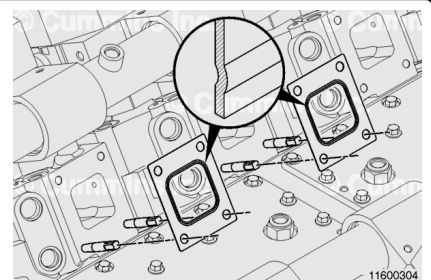


All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 1 and 2, or left bank cylinder heads number 7 and 8.

Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.



Place the exhaust manifold section onto the centering dowels.

Install the capscrews and spacers into the upper holes and center the manifold section.

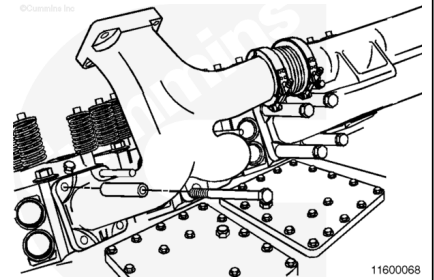
Tighten the upper capscrews.

Torque Value: 10 n•m [89 in-lb]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

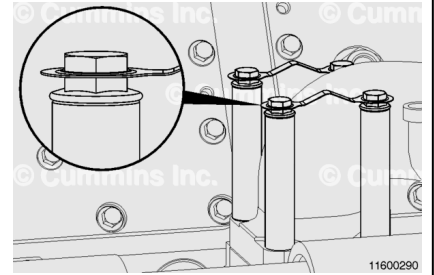
Tighten all eight capscrews in the exhaust manifold section.

Torque Value: 47 n•m [35 ft-lb]



⚠ CAUTION ⚠

Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.



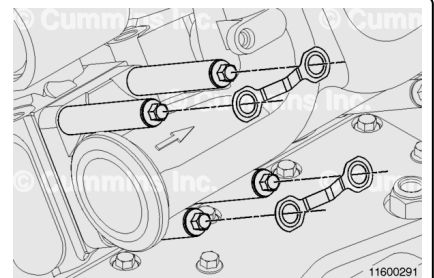
Note : Install the lockplate approximately one half of the height of the capscrew head. Installing the lockplate until it bottoms can cause it to break.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

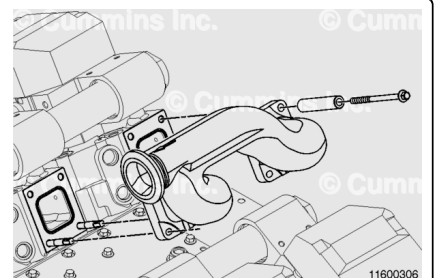
Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.



All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 7 and 8, or left bank cylinder heads number 1 and 2.



Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.

Place the exhaust manifold section onto the centering dowels.

Install the capscrews and spacers into the upper holes and center the manifold section.

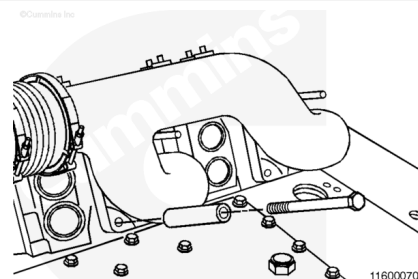
Tighten the upper capscrews.

Torque Value: 10 n•m [89 in-lb]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

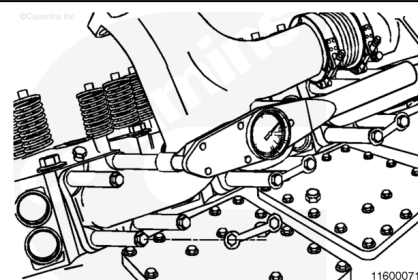
Tighten all eight capscrews in the exhaust manifold section.

Torque Value: 47 n•m [35 ft-lb]



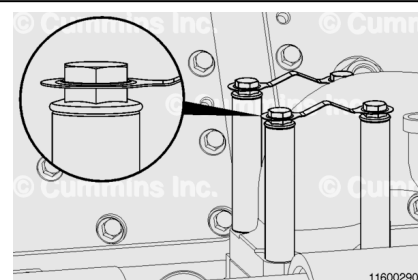
Tighten all exhaust manifold mounting capscrews in the exhaust manifold crossover sections that were **not** tightened earlier in the installation procedure.

Torque Value: 47 n•m [35 ft-lb]



⚠ CAUTION ⚠

Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.



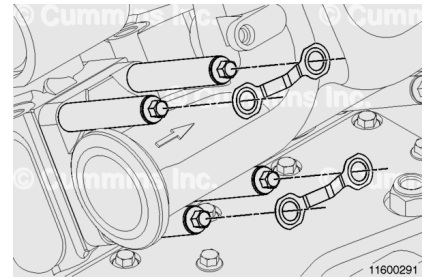
Note : Install the lockplate approximately one half of the height of the capscrew head. Installing the lockplate until it bottoms can cause it to break.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

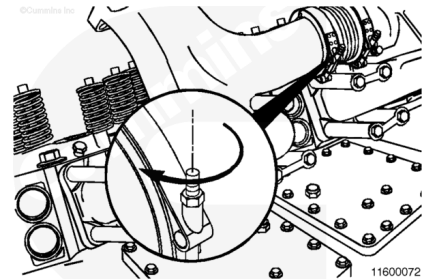
Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.



Make sure all the bellows clamps are oriented with the clamp screw toward the opposite bank of the engine with the screw pointing up. This allows accessibility after complete assembly.

Tighten all the exhaust manifold bellows clamps.

Note : There are two types of V-band clamps on the QSK45 and QSK60. They can be identified by the nut size.



Torque Value:

8mm (5/16 inch): 17 n·m [150 in-lb]

Torque Value:

6mm (1/4 inch): 10 n·m [89 in-lb]

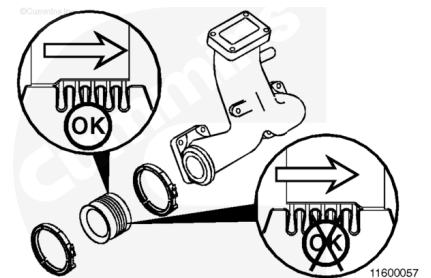
QSK60 Marine Applications

The exhaust manifold **must** be assembled as it is being installed.

Do **not** tighten the bellows clamps until all the exhaust manifolds are installed.

Apply anti-sieze compound, Part Number 3824879, to the exhaust mounting capscrews.

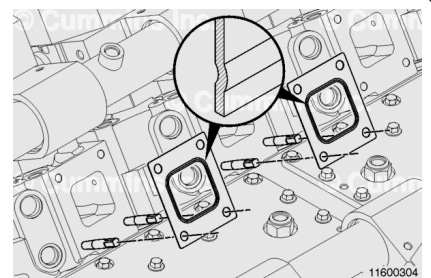
The exhaust manifold bellows are directional. Make sure they are installed in the proper manner when installing the exhaust manifold.



All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 5 and 6, or left bank cylinder heads number 3 and 4.



Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.

Place the exhaust manifold section onto the centering dowels.

Install the capscrews and spacers into the upper holes and center the manifold section.

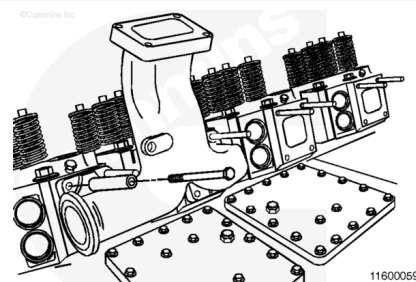
Tighten the upper capscrews.

Torque Value: 10 n•m [89 in-lb]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

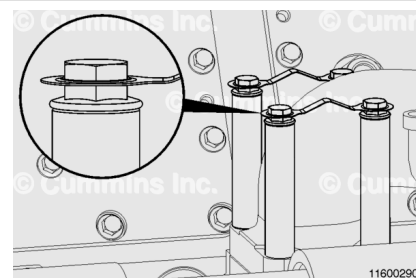
Tighten all eight capscrews in the exhaust manifold section.

Torque Value: 47 n•m [35 ft-lb]



⚠ CAUTION ⚠

Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.



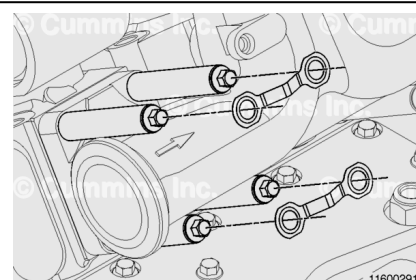
Note : When installing the lockplate over the capscrew head, install the lockplate approximately one half of the height. Installing the lockplate until it bottoms can cause it to break.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.

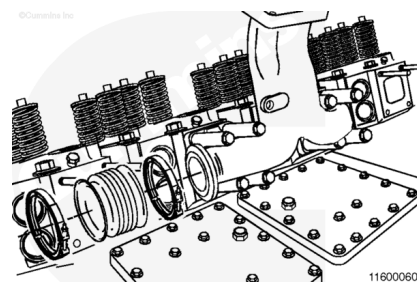


Install the bellows gasket onto each end of the bellows.

Install the bellows in the proper direction

All of the bellows clamps **must** be installed with the clamp screw toward the opposite bank of the engine, with the screw pointing up.

Loosely install the bellows clamp and bellows on the manifold section just installed.

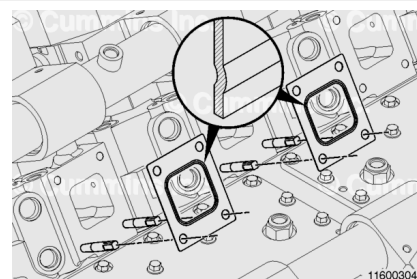


All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 3 and 4, or left bank cylinder heads number 5 and 6.

Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.



Place the exhaust manifold section onto the centering dowels.

Install the capscrews and spacers into the upper holes and center the manifold section.

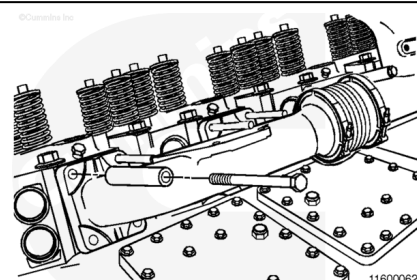
Tighten the upper capscrews.

Torque Value: 10 n•m [89 in-lb]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

Tighten all eight capscrews in the exhaust manifold section.

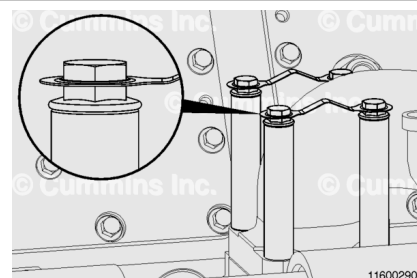
Torque Value: 47 n•m [35 ft-lb]



⚠ CAUTION ⚠

Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.

Note : When installing the lockplate over the capscrew head, install the lockplate approximately one half of the height. Installing the lockplate until it bottoms can cause it to break.

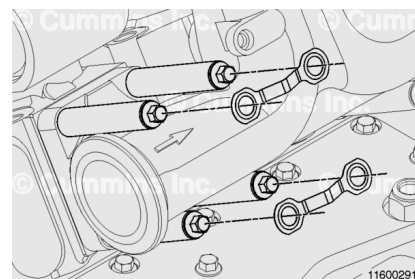


The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

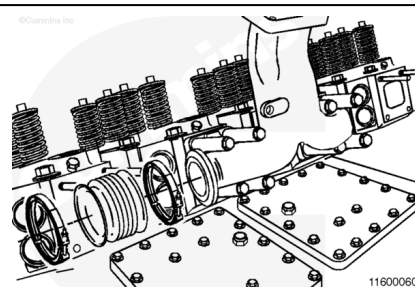
Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.



Install the bellows clamp.

Tighten the marine application bellows clamps.

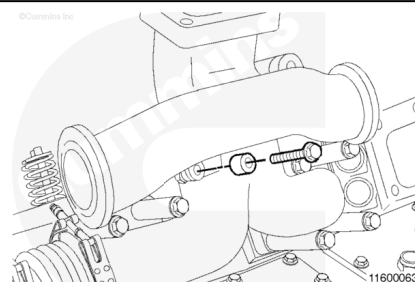
Torque Value: 17 n•m [150 in-lb]



Ring dowels are used to locate and position the crossover section.

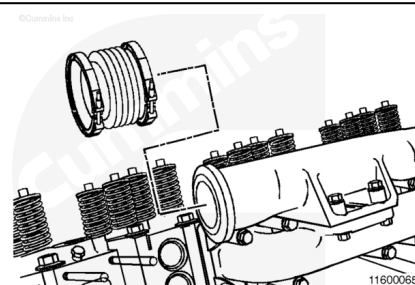
Position the crossover exhaust manifold section to the exhaust manifold section for right bank cylinders number 7 and 8, or left bank cylinders number 1 and 2.

Do **not** tighten the mounting capscrews.



Install the bellows gasket onto each end of the bellows.

Loosely install the bellows clamps and bellows to the ends of the upper exhaust manifold section.

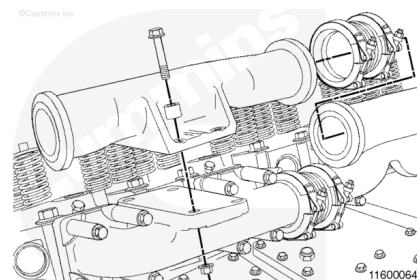


Position the crossover exhaust manifold section on top of the exhaust manifold section for right bank cylinders number 3 and 4, or left bank number 5 and 6.

Do **not** tighten the mounting capscrews.

Loosely install the bellows clamps and bellows to the upper exhaust manifold section mounted to the turbocharger mounting exhaust manifold section for right-bank cylinders number 5 and 6, left-bank cylinders number 3 and 4.

Do **not** tighten the bellows clamps.

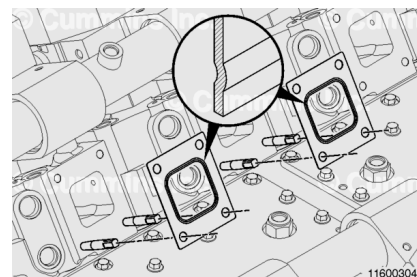


All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 1 and 2, or left bank cylinder heads number 7 and 8.

Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.



Place the exhaust manifold section onto the centering dowels.

Install the capscrews and spacers into the upper holes and center the manifold section.

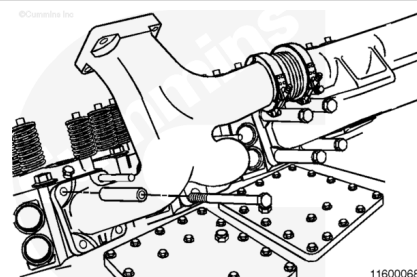
Tighten the upper capscrews.

Torque Value: 10 n•m [89 in-lb]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

Tighten all eight capscrews in the exhaust manifold section.

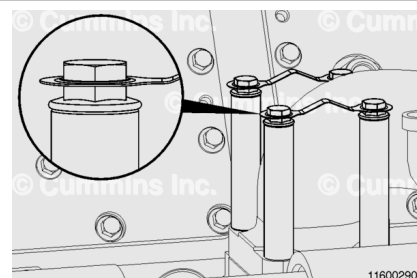
Torque Value: 47 n•m [35 ft-lb]



⚠ CAUTION ⚠

Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.

Note : Install the lockplate approximately one half of the height of the capscrew head. Installing the lockplate until it bottoms can cause it to break.

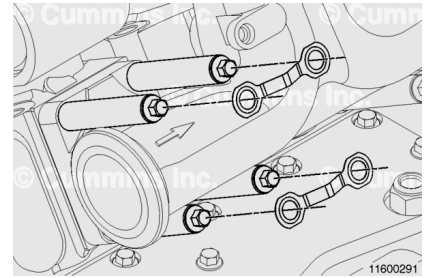


The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.

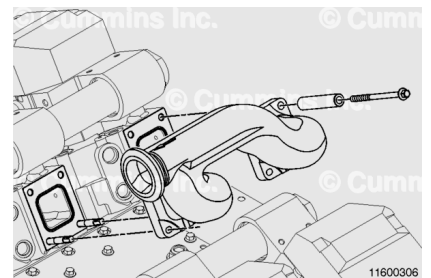


All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 7 and 8, or left bank cylinder heads number 1 and 2.

Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.



Place the exhaust manifold section onto the centering dowels.

Install the capscrews and spacers into the upper holes and center the manifold section.

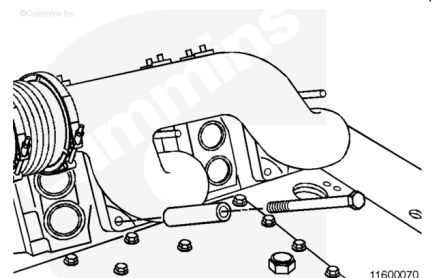
Tighten the upper capscrews.

Torque Value: 10 n•m [89 in-lb]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

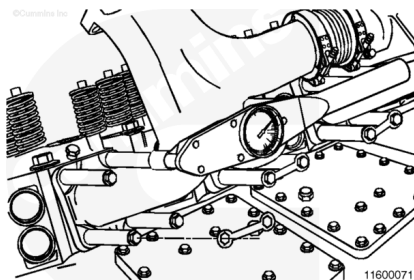
Tighten all eight capscrews in the exhaust manifold section.

Torque Value: 47 n•m [35 ft-lb]



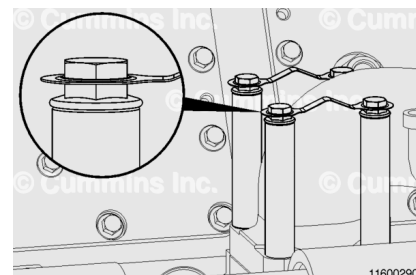
Tighten all exhaust manifold mounting capscrews in the exhaust manifold crossover sections that were **not** tightened earlier in the installation procedure.

Torque Value: 47 n•m [35 ft-lb]



⚠ CAUTION ⚠

Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.



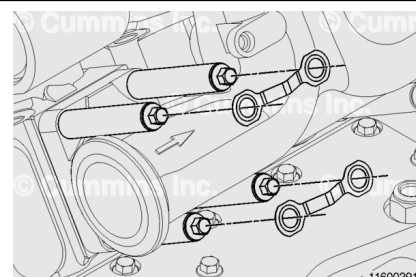
Note : Install the lockplate approximately one half of the height of the capscrew head. Installing the lockplate until it bottoms can cause it to break.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

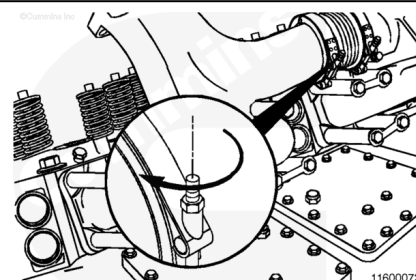
Note : For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.



Make sure all the bellows clamps are oriented with the clamp screw toward the opposite bank of the engine with the screw pointing up. This allows accessibility after complete assembly.

Tighten all the marine application bellows clamps.

Torque Value: 17 n•m [150 in-lb]

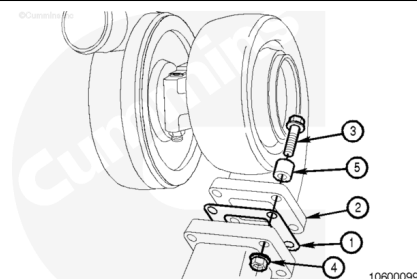


Finishing Steps

QSK45 Engines

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



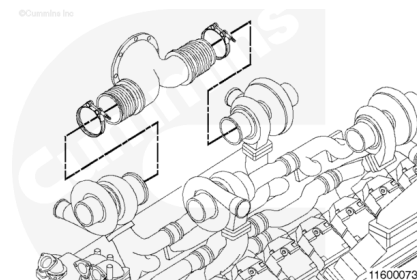
The QSK45 engine exhaust manifold is designed so that all six cylinders on each bank are connected to a single common outlet located near the center between number 3 and number 4.

Install the two turbochargers. Refer to Procedure 010-034 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>)

QSK60 Single Stage Industrial and Power Generation

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

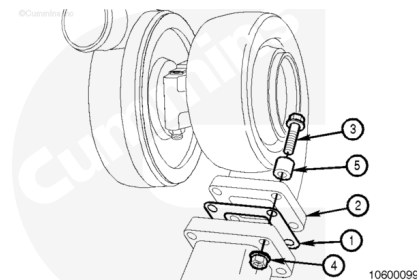


The QSK60 engine exhaust manifold is designed as a pulse-style with twin high-pressure turbocharging on each bank. Each turbocharger serves four cylinders. The turbocharger exhaust outlet feeds into a main exhaust outlet for each bank.

Install the exhaust connection and the four clamps.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the four turbochargers. Refer to Procedure 010-034 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>)

QSK60 Marine Applications

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



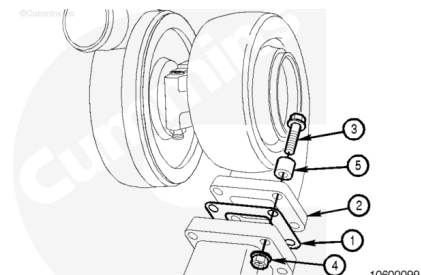
ck800wa

- Install the exhaust manifold heat shields. Refer to Procedure 011-015 in Section 11. (</qs3/pubsys2/xml/en/procedures/56/56-011-015-tr.html>)
- Install the four turbochargers. Refer to Procedure 010-034 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html>)
- Install the exhaust connection and the four clamps.
- Install the turbocharger heat shields. Refer to Procedure 010-076 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html>)
- Install the turbocharger inlet connections. Refer to Procedure 010-130 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-130-tr.html>)
- Install the aftercooler heat shields. Refer to Procedure 010-129 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html>)

QSK60 Two Stage

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



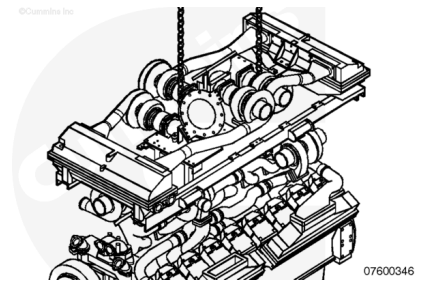
10600099

Install the high-pressure turbochargers. Refer to Procedure 010-034 in Section 10.

(/qs3/pubsys2/xml/en/procedures/56/56-010-034-tr.html)

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the two-stage platform. Refer to Procedure 010-082 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-082-tr.html)